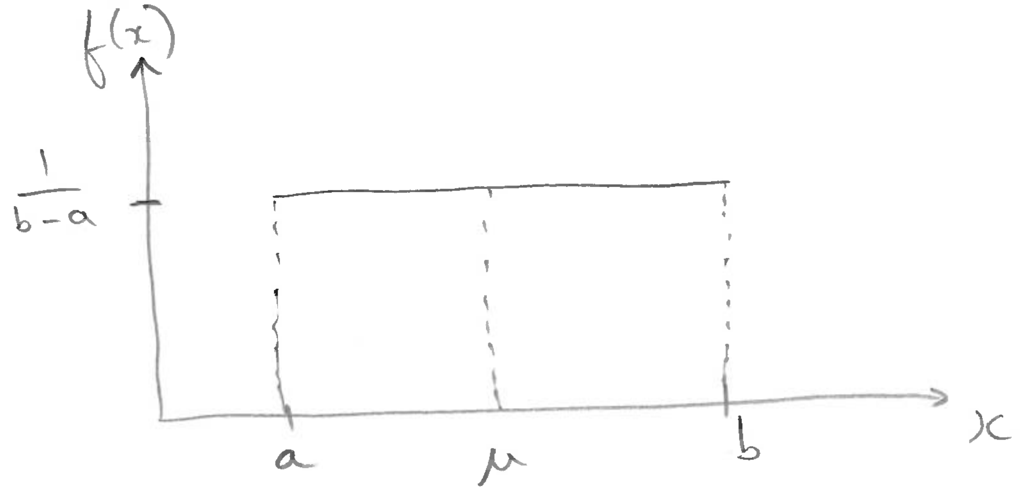
Example: Uniform distribution

The density function:

$$f(x) = \frac{1}{b-a} \quad \text{for } a \le x \le b$$

Graphically:



Properties:

$$P(X < x) = \frac{1}{b-a}(x-a) \Rightarrow \text{height} \times \text{width}$$

$$P(X > x) = \frac{1}{b-a}(b-x)$$

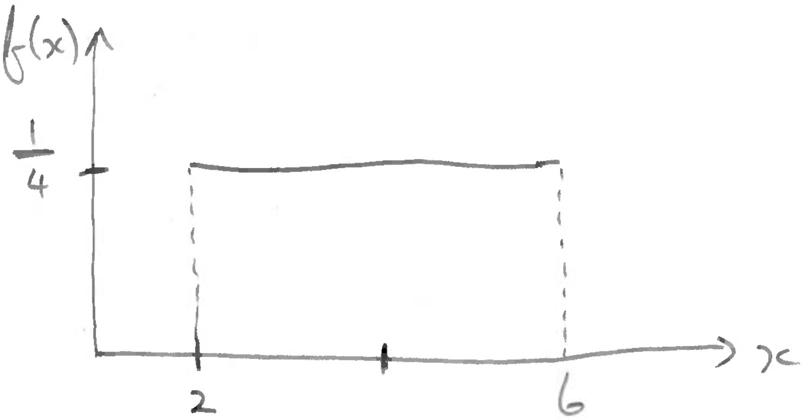
$$P(x_1 < X < x_2) = \frac{1}{b-a}(x_2 - x_1)$$

$$E(X) = \frac{a+b}{2} = \mu \text{ and } Var(X) = \frac{(b-a)^2}{12} = \sigma^2$$

Shorthand:

$$X \sim Uniform(a, b)$$

Let $X \sim \text{Uniform}(a=2,\ b=6)$



Calculate:

(a) $P(X<5)$

(b) $P(X>3)$

(c) $P(3<X<5)$

(d) $P(X<5 \mid X>3)$

(e) $P(X>3 \cup X>5)$

(f) The 80th percentile

(h) $Q_1,\ Q_2,\ Q_3$

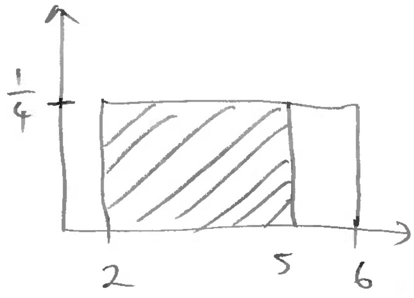
(i) $E(X)$

(j) $Var(X)$

Solutions:

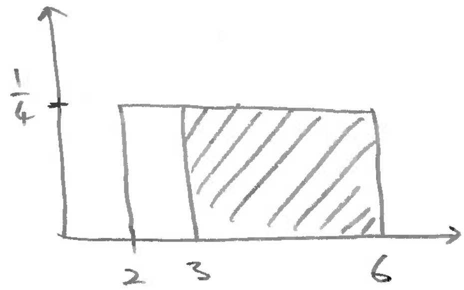
(a) $P(X<5) = \frac{1}{4}(5-2)$

$= \underline{0.75}$



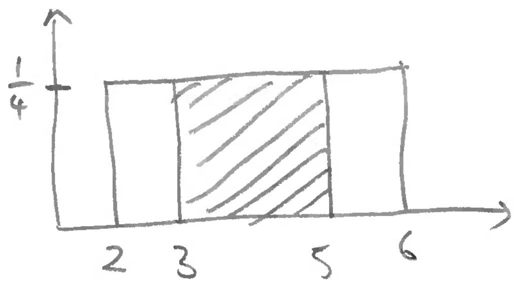
(b) $P(X>3) = \frac{1}{4}(6-3)$

$= \underline{0.75}$



(c) $P(3<X<5) = \frac{1}{4}(5-3)$

$= \underline{0.5}$



(d) $$P(X<5 \mid X>3) = \frac{P(X<5 \cap X>3)}{P(X>3)}$$

$$= \frac{P(3<X<5)}{P(X>3)}$$

$$= \frac{\frac{1}{4}(5-3)}{1/4\,(6-3)}$$

$$= \underline{0.6667}$$

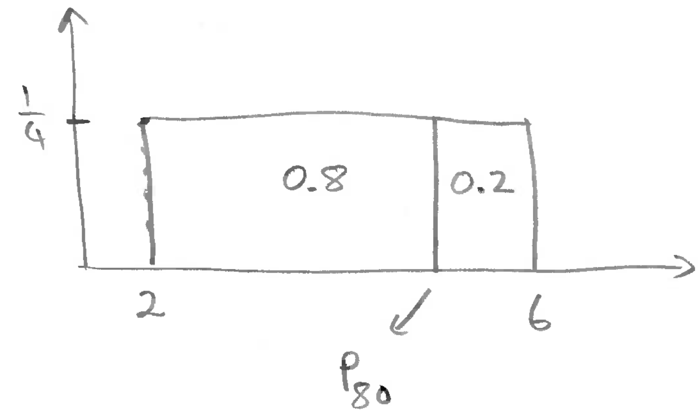
(e) $P(X>3 \cup X>5) = P(X>3) + P(X>5) - P(X>3 \cap X>5)$

$= P(X>3) + P(X>5) - P(X>5)$

$= P(X>3)$

$= \frac{1}{4}(6-3) = \underline{0.75}$

(f)



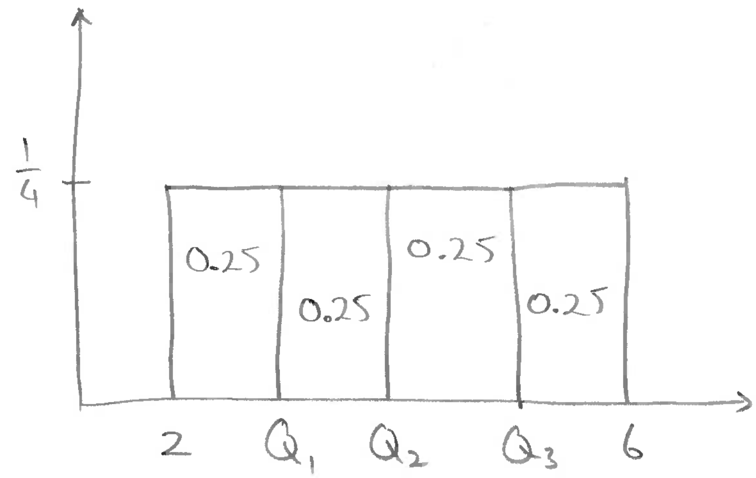
$P_{80} \Rightarrow \quad P(X<x) = 0.8$

$\frac{1}{4}(x-2) = 0.8$

$x-2 = 3.2$

$\therefore x = \underline{5.2}$

(h) Q_1, Q_2 and Q_3



$Q_1 \Rightarrow \quad P(X < x) = 0.25$

$$\frac{1}{4}(x-2) = 0.25$$

$$x - 2 = 1$$

$$\therefore x = \underline{3}$$

$Q_2 \Rightarrow \quad P(X < x) = 0.5$

$$\frac{1}{4}(x-2) = 0.5$$

$$\therefore x = \underline{4}$$

$Q_3 \Rightarrow \quad P(X < x) = 0.75$

$$\frac{1}{4}(x-2) = 0.75$$

$$\therefore x = \underline{5}$$

(i) $E(X) = \frac{a+b}{2}$

$= \frac{2+6}{2} = 4 \Rightarrow \mu$

(j) $Var(X) = \frac{(b-a)^2}{12}$

$= \frac{(6-2)^2}{12} = 1.3333 \Rightarrow \sigma^2$